

SatQuery-Edge

Offline Satellite Intelligence Investigation Report
SIH 26167 · ISRO · Prototype · Offline Mode

Investigation ID	SQE-TEST-001
Timestamp	2026-09-02 11:26:38
Intent	bi_temporal_change
Tools Used	change, grounding, geospatial

Query

Test query

Final Answer

Test answer

Evidence Score

Component	Code	Value	Weight
Model Agreement	M	0.894	0.30
Sensor Reliability	S	0.900	0.20
Spatial Consistency	P	0.855	0.20
Temporal Consistency	T	0.940	0.15
Input Quality	I	0.534	0.15
FINAL SCORE		84.0 / 100	

Validation

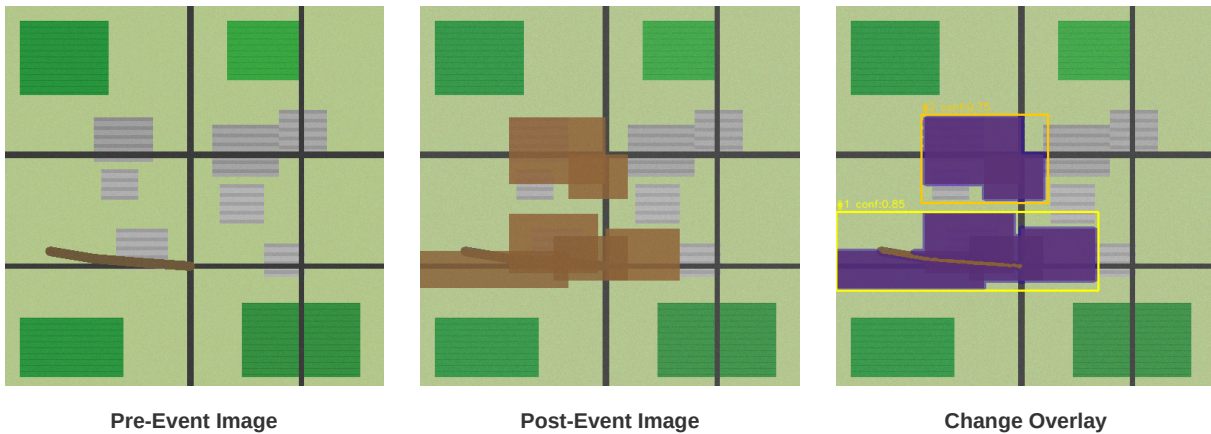
Status: **PASS** ✓

- 2 image(s) detected — count OK.
- Image 1: Quality OK (mean=162, std=40.1).
- Image 2: Quality OK (mean=158, std=37.7).
- Image pair: identical dimensions — perfect alignment.

Detected Regions & Approximate Coordinates

#	Label	Confidence	Area (km²)	Demo Georeferenced
1	candidate changed region	0.85	18.493	Yes
2	candidate changed region	0.76	10.029	Yes

Key Visual Evidence



Specialist Tool Outputs

CHANGE — SUCCESS — 2 region(s)

Detected 2 significant candidate changed region(s).

Largest region:

Approximate change area: 10.2% of image.

Confidence: 0.85

Center: (0.37, 0.66) (normalized)

Total changed pixels: ~16.4% of scene.

Interpretation:

Candidate newly inundated / possible flooded regions where spectral properties changed significantly between observations.

New water-like spectral signatures appear over

GROUNDING — SUCCESS — 0 region(s)

No candidate regions found matching 'water'.

GEOSPATIAL — SUCCESS — 0 region(s)

Geospatial conversion complete.

Converted 2 region(s) to geographic coordinates.

Total approximate area: 28.52 km².

■ Approximate demo georeferencing applied.

Demo center: (20.296°N, 85.824°E)

Coordinates are NOT derived from real GeoTIFF metadata.

Execution Trace

Tool Versions

Component	Version
SatQuery-Edge Prototype	1.0.0-SIH26167
Python OpenCV	4.11.0
NumPy	1.26.4
Streamlit	1.62.0

Report Engine	ReportLab PDF
Mode	OFFLINE — No external APIs

NOTE: This prototype uses lightweight local computer vision and synthetic/demo geospatial observations where applicable. Results are intended for demonstration purposes only and are NOT scientifically validated remote-sensing products. No cloud APIs, external inference services, or internet connectivity is used. All processing is fully offline. —
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